

LAPORAN VECTOR DATABASE & INDEXING

Corpus RAG AMIKOM · Versi 1

Database lokal persisten dengan FAISS IndexIDMap2(IndexFlatIP) dan SQLite metadata store.

Scope	Embedding Gate	Vector DB Gate	Ready for Retrieval
FROZEN	PASS	PASS	YES

Dokumen	Nilai
Database name	rag_amikom_v1
Versi FAISS	1.14.3
Dimensi / jumlah	384 / 306
Waktu pembuatan	2026-07-27T12:18:55.955+07:00
Reviewer project	Mikhail Satria
Persetujuan download mode	26 Juli 2026 · 04.01 WIB

Tahap ini berhenti pada Vector Database Gate. Tidak ada retrieval evaluation, penetapan top-k final, baseline RAG, generation test, implementasi multi-agent, atau antarmuka aplikasi.

1. Ringkasan Eksekutif

Sebanyak 306 vector embedding berdimensi 384 berhasil dimasukkan tanpa pembuatan ulang embedding. Vector disimpan pada indeks exact FAISS dan hubungan ke chunk text serta metadata disimpan pada SQLite. Indeks ditutup, dibuka kembali, dan diverifikasi tetap berisi 306 record.

Indikator	Hasil	Bukti ringkas
Indexed vector	306	Sama dengan expected 306
Vector dimension	384	Konsisten pada input dan database
Namespace	5	243 / 3 / 25 / 10 / 25
Checkpoint	10	Seluruh batch COMPLETE
Reload	PASS	FAISS ntotal = SQLite count = 306
Smoke test	7/7 PASS	Lookup dan filter metadata saja
QA	18/18 PASS	Q01-Q18 tanpa kegagalan

Keputusan Gate

Vector Database Gate = PASS dan Ready for Retrieval Testing = YES. Retrieval Testing Executed tetap NO.

2. Tujuan dan Batas Tahap

- Membuat vector database lokal dan persisten.
- Memasukkan tepat 306 vector beserta chunk text dan metadata.
- Menjaga isolasi namespace/lifecycle melalui metadata filter terindeks.
- Membuktikan database dapat dibuka kembali dan siap untuk tahap berikutnya.

Yang tidak dilakukan

Tidak ada similarity-search evaluation, pengukuran precision/recall, top-k tuning, baseline RAG, generation test, multi-agent, maupun UI.

3. Input Wajib dan Preflight

Enam artefak input dibaca dari versi yang disahkan pada tahap embedding. Validasi preflight menghentikan proses bila jumlah, urutan, isi, atau hash tidak cocok.

Input	Bytes	SHA-256	Status
Embedding_Corpus_RAG_AMIKOM_V1.zip	1330562	f72a7d7f81c24099de4c039bcd44611a12f00bd2c23675e00b3d97a7276e472d	VERIFIED
Manifest_Embedding_RAG_AMIKOM_V1.xlsx	152329	e5973f317d1c4d356d65bd7689bda6cde104e41b26a00b6fdf391e eee3edd8a6	VERIFIED
Embedding_Config_RAG_AMIKOM_V1.json	3121	df0b999d321321c94daa210fb3316ba0465e744dbde37d24a88212 5b1a8e3ede	VERIFIED
Laporan_Embedding_RAG_AMIKOM_V1.pdf	97065	d2eb580cf6272e8fc39d84af0794a8589f705d9fca9359bb2973da 8d70edc4c9	VERIFIED
README_Embedding_RAG_AMIKOM_V1.md	3650	ff3d1ba5f4afc7b133f2680574f87ced62b6491343be526ac4c128 3146f8b0a2	VERIFIED
Chunk_Corpus_RAG_AMIKOM_V1.jsonl	1093843	9bc4147b2872db0dba4ff46b0d1050662f61975b672555212ddabd f5b68e174e	VERIFIED

Pemeriksaan preflight	Status
P01	PASS
P02	PASS
P03	PASS
P04	PASS
P05	PASS
P06	PASS
P07	PASS
P08	PASS
P09	PASS
P10	PASS
P11	PASS
P12	PASS
P13	PASS
P14	PASS
P15	PASS
P16	PASS

Hasil: Embedding Gate PASS; embeddings.npy 306 × 384; 306 metadata; 306 chunk text; vector_index 0–305 berurutan; vector_id/chunk_id unik; tanpa NaN, Infinity, atau zero vector; input hash tidak berubah.

4. Pemilihan Database dan Arsitektur

Chroma persistent local tidak tersedia pada environment saat eksekusi. Sesuai urutan prioritas, digunakan fallback FAISS 1.14.3 dengan SQLite 3.50.4 sebagai metadata store persisten.

Parameter	Nilai
database_name	rag_amikom_v1
database_type	FAISS IndexIDMap2(IndexFlatIP) + SQLite metadata store
library_version	FAISS 1.14.3; SQLite 3.50.4
persistence_path	01_database/
distance_metric	cosine_via_normalized_inner_product
index_type	IndexIDMap2(IndexFlatIP)
environment	Linux-6.12.13-x86_64-with-glibc2.39; CPU; Python 3.12.13
created_at_wib	2026-07-27T12:18:55.955+07:00

Cosine similarity

Vector embedding sudah L2-normalized. FAISS IndexFlatIP menggunakan inner product; untuk vector normalized hasilnya ekuivalen dengan cosine similarity. ID int64 FAISS menggunakan vector_index 0-305, sedangkan vector_id stabil dari tahap embedding tetap dipertahankan.

5. Persistence dan Relasi Record

Komponen	Peran
01_database/faiss.index	Vector float32, dimensi 384, ID deterministik vector_index.
01_database/metadata.sqlite	Chunk text, seluruh metadata wajib, filter namespace/lifecycle.
03_metadata/indexed_records.jsonl	Snapshot hubungan record untuk audit dan recovery.
04_checkpoint/batch_checkpoints	10 checkpoint JSON + NPY per batch.

Relasi yang dipertahankan: vector ↔ vector_index ↔ vector_id ↔ chunk_id ↔ chunk_text ↔ metadata sumber. Tidak ada UUID acak dan tidak ada record setengah lengkap.

6. Proses Indexing Deterministik

Urutan kerja: membuka embeddings.npy; membaca embedding_metadata.jsonl; menghubungkan chunk_text via chunk_id; memvalidasi vector_index; insert SQLite dan FAISS per batch; menulis checkpoint; persist; tutup; buka kembali; QA.

Batch	Index	Jumlah	Status	Output hash
1	0-31	32	COMPLETE	e32170359bda88f24ed0b3fb24b8f48850ec6ba863329bb4b04ce6d5486b7991
2	32-63	32	COMPLETE	69f924ffc42727ecbc603a80616f8e169bb77d008e762111fded78c1201dae28
3	64-95	32	COMPLETE	d68087e7425cd6b72fef22ec835568c3df0f8da499e4aa1c3650ce99d22dc9f9
4	96-127	32	COMPLETE	a88221b5db34a30882c2a18766be760f2f4e6807e8f1d97796d44f66665376a9
5	128-159	32	COMPLETE	2b7997ffc45062d603458188d44bb2c203601e62f7ee6f98af209d52d5f2e9c9
6	160-191	32	COMPLETE	e12c0711cf4fe307d01074a2891294e2a8a35a7d569055114676fdd01c50c65d
7	192-223	32	COMPLETE	0fa76b647117bfb6ffdb90f5ab343b29d0675a2de93b36264bd77715c3ca8128
8	224-255	32	COMPLETE	f255be492642d4d7d53a3fadf806d36251481122bfb64450ff3cdc2c9a654dda
9	256-287	32	COMPLETE	21823310aa86b590ab32df69f534419daf6c53ddb137737b81eee865ea520483
10	288-305	18	COMPLETE	f1792ee5e88338ec625ea1f5e9740f2610eccc82fd215c6da58688757e92c1b1

Resume test

Status PASS: 10/10 batch COMPLETE terdeteksi dan dilewati tanpa reinsertion.

7. Distribusi Namespace dan Lifecycle

Namespace	Record	Active	Historical
active_academic	243	243	0
active_administration	25	25	0
active_dynamic_schedule	10	10	0
active_schedule	3	3	0
archive_schedule	25	0	25

Lifecycle	Record	Active	Historical
ACTIVE	271	271	0
ACTIVE_DYNAMIC	10	10	0
ARCHIVE	25	0	25

ACTIVE_DYNAMIC: 10/10 record memiliki ttl_days=1, live_check_required=true, dan snapshot_at_wib.
ARCHIVE: 25/25 record historical_only=true dan active_retrieval_allowed=false.

8. Kelengkapan Metadata

Field	Record	Nonblank	Missing	Status
vector_id	306	306	0	PASS
vector_index	306	306	0	PASS
chunk_id	306	306	0	PASS
chunk_text	306	306	0	PASS
source_id	306	306	0	PASS
document_id	306	306	0	PASS
title	306	306	0	PASS
locator	306	306	0	PASS
agent_namespace	306	306	0	PASS
retrieval_namespace	306	306	0	PASS
lifecycle_status	306	306	0	PASS
academic_year	306	306	0	PASS
semester	306	306	0	PASS
cohort	306	306	0	PASS
freshness_status	306	306	0	PASS
ttl_days	306	306	0	PASS
live_check_required	306	306	0	PASS
snapshot_at_wib	306	306	0	PASS
historical_only	306	306	0	PASS
active_retrieval_allowed	306	306	0	PASS
source_priority	306	306	0	PASS
chunk_hash	306	306	0	PASS
model_name	306	306	0	PASS
embedding_version	306	306	0	PASS

Nilai eksplisit NOT_AVAILABLE/NOT_APPLICABLE dianggap nilai metadata, bukan cell kosong. Seluruh chunk_text tersedia dan tidak ada metadata wajib yang hilang.

9. Reload dan Sample Record

Uji	Hasil
Reload FAISS	PASS
FAISS ntotal	306
SQLite record count	306
Dimension	384
Exact reconstructed vector	PASS
Sample vector_id	VEC-E5S-CH-ACA-A01-DOC-A01-ACADEMIC-POLICY-0001

Persistence result

Database ditutup, dibuka ulang dari file fisik, dan jumlah record tetap 306. Sample record berhasil diambil berdasarkan vector_id.

10. Filter Smoke Test

Smoke test dibatasi pada lookup dan filter metadata. Tidak ada query embedding, ranking similarity, pengukuran retrieval, atau evaluasi pertanyaan.

ID	Test	Expected	Actual	Status	Evidence
SMK-01	Lookup exact record by vector_id	1	1	PASS	Resolved VEC-E5S-CH-ACA-A01-DOC-A01-ACADEMIC-POLICY-0001; no similarity search executed.
SMK-02	Filter namespace active_academic	243	243	PASS	Namespace metadata index returned the expected count.
SMK-03	Filter lifecycle ARCHIVE	25	25	PASS	Historical lifecycle filter returned only archive records.
SMK-04	Current filter excludes archive	281	281	PASS	Current filter contains ACTIVE and ACTIVE_DYNAMIC only.
SMK-05	Current filter contains zero archive	0	0	PASS	Archive isolation confirmed.
SMK-06	ACTIVE_DYNAMIC control filter	10	10	PASS	All dynamic records retain TTL, live-check, and snapshot.
SMK-07	Filter namespace active_schedule	3	3	PASS	Schedule namespace count matches the frozen manifest.

Hasil smoke test

7/7 PASS. Current filter menghasilkan 281 record ACTIVE/ACTIVE_DYNAMIC dan mengambil 0 archive; filter ARCHIVE menghasilkan tepat 25 record.

11. Isolasi Archive dan Dynamic

- archive_schedule dipisahkan melalui retrieval_namespace dan lifecycle_status=ARCHIVE.
- Current filter wajib active_retrieval_allowed=true AND historical_only=false.
- ACTIVE_DYNAMIC tetap aktif, tetapi TTL=1 dan live-check dipertahankan.
- CONFLICT, BLOCKED, dan CONTROL tidak ada di database vector.

12. QA Registry Q01-Q18

ID	Kriteria	Status	Bukti
Q01	Embedding input tepat 306 vector.	PASS	(306, 384)
Q02	Database berisi tepat 306 record.	PASS	306
Q03	Dimensi seluruh vector = 384.	PASS	306 records
Q04	Seluruh vector_id unik.	PASS	306
Q05	Seluruh chunk_id unik.	PASS	306
Q06	Tidak ada CONFLICT, BLOCKED, atau CONTROL.	PASS	{'ACTIVE': 271, 'ACTIVE_DYNAMIC': 10, 'ARCHIVE': 25}
Q07	Distribusi namespace sesuai 243/3/25/10/25.	PASS	{'active_academic': 243, 'active_administration': 25, 'active_dynamic_schedule': 10, 'active_schedule': 3, 'archive_schedule': 25}
Q08	Seluruh chunk_text tersedia.	PASS	306 non-empty
Q09	Seluruh metadata wajib tersedia.	PASS	24 fields PASS
Q10	Vector dan metadata tidak tertukar.	PASS	vector_index/vector_id/chunk_id order matched
Q11	ACTIVE_DYNAMIC memiliki TTL dan live-check.	PASS	{'test_id': 'SMK-06', 'test_name': 'ACTIVE_DYNAMIC control filter', 'sql_filter': "SELECT vector_id FROM vector_records WHERE lifecycle_status='ACTIVE_DYNAMIC' AND ttl_days=1 AND live_check_required=1 AND snapshot_at_wib NOT IN ('','NOT_AVAILABLE','NOT_APPLICABLE')", 'expected_count': 10, 'actual_count': 10, 'status': 'PASS', 'evidence': 'All dynamic records retain TTL, live-check, and snapshot.'}
Q12	ARCHIVE memiliki active_retrieval_allowed=false.	PASS	25/25
Q13	Database persisten dan dapat dibuka kembali.	PASS	306
Q14	Jumlah record tetap 306 setelah reload.	PASS	306
Q15	Sample record dapat diambil berdasarkan vector_id.	PASS	VEC-E5S-CH-ACA-A01-DOC-A01-ACADEMIC-POLICY-0001
Q16	Metadata filter namespace dan lifecycle berfungsi.	PASS	7/7 PASS
Q17	Hash input dan output tercatat.	PASS	{'inputs': 6, 'faiss_index': 'e7d83504bfbcc131049896f66921f4b00535dd22b0c0c74cfd25811ac69f4b7', 'metadata_sqlite': '0d1ee79ec95d0341a448218681d95b04f85a5cb61ac5942611f194592c6b4b4d'}
Q18	Corpus dan embedding asli tidak berubah.	PASS	{'Embedding_Corpus_RAG_AMIKOM_V1.zip': {'size_bytes': 1330562, 'sha256': 'f72a7d7f81c24099de4c039bcd44611a12f00bd2c23675e00b3d97a7276e472d'}, 'Manifest_Embedding_RAG_AMIKOM_V1.xlsx': {'size_bytes': 152329, 'sha256': 'e5973f317d1c4d356d65bd7689bda6cde104e41b26a00b6fdf391eeee3edd8a6'}, 'Embedding_Config_RAG_AMIKOM_V1.json': {'size_bytes': 3121, 'sha256': 'df0b999d321321c94daa210fb3316ba0465e744dbde37d24a882125b1a8e3ede'}, 'Laporan_Embedding_RAG_AMIKOM_V1.pdf': {'size_bytes': 97065, 'sha256': 'd2eb580cf6272e8fc39d84af0794a8589f705d9fca9359bb2973da8d70edc4c9'}, 'README_Embedding_RAG_AMIKOM_V1.md': {'size_bytes': 3650, 'sha256': 'ff3d1ba5f4afc7b133f2680574f87ced62b6491343be526ac4c1283146f8b0a2'}}, 'Chunk_Corpus_RAG_AMIKOM_V1.jsonl': {'size_bytes': 1093843, 'sha256': '9bc4147b2872db0dba4ff46b0d1050662f61975b67255212ddabdf5b68e174e'}}

Hasil QA

18/18 PASS; 0 FAIL. Seluruh syarat Vector Database Gate terpenuhi.

13. Hash Input dan Output Inti

Q17 dan Q18 memverifikasi hash input dan output serta memastikan Raw, Clean, Chunk, dan Embedding Corpus tidak dimodifikasi.

Artefak inti	SHA-256
FAISS index	e7d83504bfbb131049896f66921f4b00535dd22b0c0c74cfd25811ac69f4b7
SQLite metadata	0d1ee79ec95d0341a448218681d95b04f85a5cb61ac5942611f194592c6b4b4d
Indexed records JSONL	2207b6283755b35b10e9b40e0b5637d15eb4c26cc4960aa0ace1863ba6e199a5
Configuration	9c742bb4917a97de92e3341e772ac9a06d6984a6c33e9f0456adc5b8b669245c

Hash ZIP final tidak dimasukkan ke dalam ZIP itu sendiri karena akan menciptakan referensi sirkular. Nilai final dicatat pada final_package_validation.json dan vector_database_package.sha256.

14. Blocker Registry

ID	Status	Database action	Response / handoff
G01	PARTIALLY_RESOLVED_ACCEPTED_BLOCKER	10 ACTIVE_DYNAMIC records indexed with TTL=1, live_check_required=true, and snapshot preserved.	LIVE_CHECK_OR_ABSTAIN · DAAK / official academic calendar channel
G02	OPEN_ACCEPTED_BLOCKER	No synthetic agenda record indexed.	ABSTAIN · DAAK / official agenda publication channel
G04	PARTIALLY_RESOLVED	Course-code precedence metadata retained; conflict remains excluded.	ANSWER_WITH_PRECEDENCE_AND_ESCALATE_CONFLICT · Program Studi S1 Informatika
CF002	OPEN_ACCEPTED_BLOCKER	No CONFLICT record indexed.	ESCALATE · Fakultas Ilmu Komputer / DAAK
MR-A11-RELATIONS	NEEDS_MANUAL_REVIEW	No CONTROL relation record indexed; ambiguity remains answer-disabled.	ASK_CONTEXT_OR_ESCALATE · Program Studi S1 Informatika

Integritas blocker

G01, G02, G04, CF002, dan MR-A11-RELATIONS tetap dicatat sesuai status sebelumnya. Accepted blocker tidak diubah menjadi fakta sintetis.

15. Vector Database Gate

Status akhir	Nilai
Scope Status	FROZEN
Embedding Gate	PASS
Database Type	FAISS IndexIDMap2(IndexFlatIP) + SQLite metadata store
Indexed Vector	306
Expected Vector	306
Vector Dimension	384
Vector Database Status	COMPLETE
Vector Database Gate	PASS
Ready for Retrieval Testing	YES
Retrieval Testing Executed	NO
Baseline RAG Executed	NO
Ready for Multi-Agent	NO

Keputusan

Vector Database Gate PASS. Database lokal persisten siap untuk retrieval testing, namun pengujian retrieval belum dimulai pada tahap ini.

16. Referensi Teknis

- PyPI faiss-cpu: <https://pypi.org/project/faiss-cpu/>
- FAISS installation: <https://github.com/facebookresearch/faiss/blob/main/INSTALL.md>
- FAISS metric guidance: <https://github.com/facebookresearch/faiss/wiki/MetricType-and-distances>

Dokumen ini merekam keadaan pada waktu eksekusi. File database, checkpoint, manifest, log, QA, dan hash registry tersedia dalam paket ZIP yang menyertai laporan.